MILITARY SPECIFICATION SHEET

# CIRCUIT BREAKERS, MAGNETIC, LOW-POWER, SEALED, TRIP-FREE

## THREE-POLE, AUXILIARY CONTACTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the circuit breakers described herein shall consist of this specification and the latest issue of MIL-C-39019.

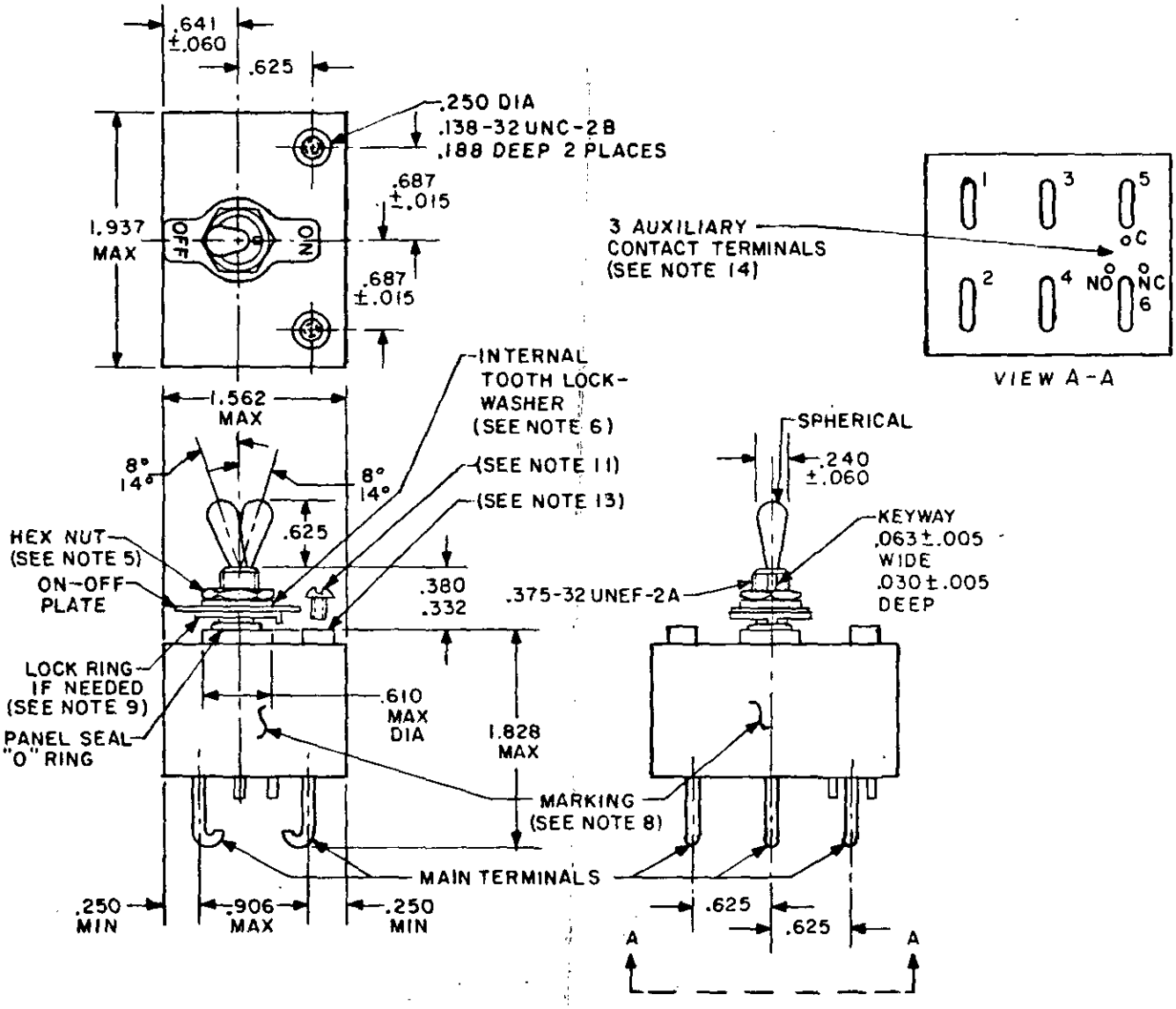
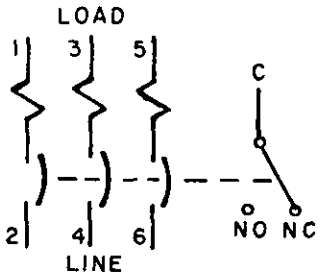



Figure 1. Circuit breaker configuration.

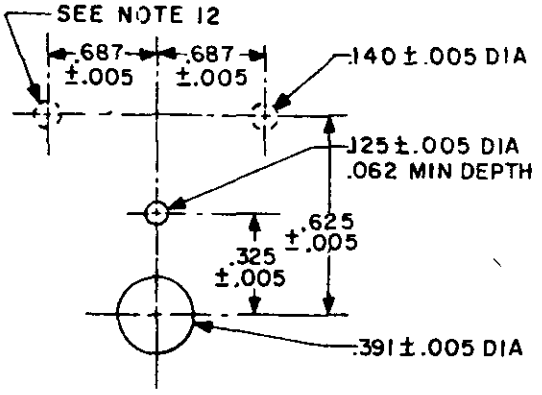
(E) denotes changes



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CIRCUIT DIAGRAM
(MAIN CONTACTS OPEN)
(SEE NOTE 10)




PANEL MTG DETAIL

| INCHES | MM | INCHES | MM |
|---|---|---|---|
| .005 | .13 | .375 | 9.53 |
| .010 | .25 | .380 | 9.65 |
| .015 | .38 | .382 | 9.70 |
| .025 | .64 | .391 | 9.93 |
| .030 | .76 | .392 | 9.96 |
| .031 | .79 | .493 | 12.52 |
| .060 | 1.52 | .500 | 12.70 |
| .063 | 1.60 | .507 | 12.88 |
| .093 | 2.36 | .610 | 15.49 |
| .125 | 3.18 | .625 | 15.88 |
| .140 | 3.56 | .641 | 16.28 |
| .188 | 4.78 | .687 | 17.45 |
| .240 | 6.10 | .906 | 23.01 |
| .325 | 8.26 | 1.562 | 39.67 |
| .332 | 8.43 | 1.828 | 46.43 |
| | | 1.937 | 49.20 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Tolerance is ±.031 unless otherwise specified.
4. Envelope design optional.
5. Hex mounting nut .375-32UNEF-2B thread, .500 ±.010 across flats, .093 ±.005 thick, brass nickel plated, QQ-N-290 nonglare, or stainless steel.
6. Internal tooth lockwasher, .507/.493 O.D., .392/.382 I.D., .025 ±.005 thick, stainless steel.
7. The effective bushing thread length (not including mounting hardware) is .280 minimum.
8. Marking may appear on any surface except the mounting surface.
9. Lock ring not required if on-off plate has locking tab.
10. Numerical marking optional on circuit diagram.
11. Sealscrew, .138-32UNC-2A x .312 long, stainless steel No. 18-8 (type 347) with O-Ring-Buna N Durometer 60-70. Possible source, APM Hexseal Corp.
12. Use of panel screws will enhance circuit breaker shock and vibration characteristics
13. Threaded inserts exceed height of bushing by .005 to .015.
14. Auxiliary contact terminals shall be located on the back side of the circuit breaker (same surface as main terminals), but otherwise their location, style, and design shall be optional, consistent with dimensional or other specified requirements. Auxiliary contact terminals shall accomodate two No. 20 AWG wires.

Figure 1. Circuit breaker configuration - Continued.



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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: For reference only, approximately 150 grams.

Mounting hardware: See figure 1.

Terminals: See figure 1.

Voltage and frequency rating: 50 Vdc, maximum and 240 Vac, maximum, at 60 and 400 Hz.

High inrush: 800 percent at 60 Hz and dc, 1400 percent at 400 Hz (see table I.)

Current rating: See table I.

Circuit diagram: See figure 1.

Time delay: See table I.

Resistance or impedance: See table I.

Part number: M39019/06-(dash number taken from table I) (e.g., M39019/06-200).

Cross reference: See table II.



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TABLE I. Circuit breaker dash numbers and applicable characteristics.

| Dash number | | Current rating (amps) | Time delay | Resistance or impedance (ohms - max) 1/ | | | Dash number | | Current rating (amps) | Time delay | Resistance or impedance (ohms - max) 1/ | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Standard delay | High inrush | | | DC | 60 Hz | 400 Hz | Standard delay | High inrush | | | DC | 60 Hz | 400 Hz |
| 200 | 300 | 0.05 | A | 680.0 | 690.0 | 710.0 | 230 | 320 | 5.0 | A | 0.061 | 0.063 | 0.072 |
| 201 | 301 | 0.05 | B | 680.0 | 690.0 | 710.0 | 231 | 321 | 5.0 | B | 0.061 | 0.063 | 0.072 |
| 202 | --- | 0.05 | C | 680.0 | 690.0 | 710.0 | 232 | --- | 5.0 | C | 0.061 | 0.063 | 0.072 |
| 203 | 302 | 0.1 | A | 150.0 | 170.0 | 180.0 | 233 | 322 | 6.0 | A | 0.042 | 0.043 | 0.050 |
| 204 | 303 | 0.1 | B | 150.0 | 170.0 | 180.0 | 234 | 323 | 6.0 | B | 0.042 | 0.043 | 0.050 |
| 205 | --- | 0.1 | C | 150.0 | 170.0 | 180.0 | 235 | --- | 6.0 | C | 0.042 | 0.043 | 0.050 |
| 206 | 304 | 0.25 | A | 20.0 | 26.0 | 27.0 | 236 | 324 | 7.0 | A | 0.036 | 0.036 | 0.040 |
| 207 | 305 | 0.25 | B | 20.0 | 26.0 | 27.0 | 237 | 325 | 7.0 | B | 0.036 | 0.036 | 0.040 |
| 208 | --- | 0.25 | C | 20.0 | 26.0 | 27.0 | 238 | --- | 7.0 | C | 0.036 | 0.036 | 0.040 |
| 209 | 306 | 0.5 | A | 5.4 | 6.0 | 6.6 | 239 | 326 | 7.5 | A | 0.031 | 0.031 | 0.038 |
| 210 | 307 | 0.5 | B | 5.4 | 6.0 | 6.6 | 240 | 327 | 7.5 | B | 0.031 | 0.031 | 0.038 |
| 211 | --- | 0.5 | C | 5.4 | 6.0 | 6.6 | 241 | --- | 7.5 | C | 0.031 | 0.031 | 0.038 |
| 212 | 308 | 0.75 | A | 2.5 | 2.7 | 2.8 | 242 | 328 | 8.0 | A | 0.027 | 0.028 | 0.035 |
| 213 | 309 | 0.75 | B | 2.5 | 2.7 | 2.8 | 243 | 329 | 8.0 | B | 0.027 | 0.028 | 0.035 |
| 214 | --- | 0.75 | C | 2.5 | 2.7 | 2.8 | 244 | --- | 8.0 | C | 0.027 | 0.028 | 0.035 |
| 215 | 310 | 1.0 | A | 1.35 | 1.5 | 1.61 | 245 | 330 | 9.0 | A | 0.022 | 0.022 | 0.028 |
| 216 | 311 | 1.0 | B | 1.35 | 1.5 | 1.61 | 246 | 331 | 9.0 | B | 0.022 | 0.022 | 0.028 |
| 217 | --- | 1.0 | C | 1.35 | 1.5 | 1.61 | 247 | --- | 9.0 | C | 0.022 | 0.022 | 0.028 |
| 218 | 312 | 2.0 | A | 0.40 | 0.40 | 0.50 | 248 | 332 | 10.0 | A | 0.018 | 0.021 | 0.024 |
| 219 | 313 | 2.0 | B | 0.40 | 0.40 | 0.50 | 249 | 333 | 10.0 | B | 0.018 | 0.021 | 0.024 |
| 220 | --- | 2.0 | C | 0.40 | 0.40 | 0.50 | 250 | --- | 10.0 | C | 0.018 | 0.021 | 0.024 |
| 221 | 314 | 2.5 | A | 0.25 | 0.25 | 0.27 | 251 | 334 | 12.5 | A | 0.012 | 0.013 | 0.015 |
| 222 | 315 | 2.5 | B | 0.25 | 0.25 | 0.27 | 252 | 335 | 12.5 | B | 0.012 | 0.013 | 0.015 |
| 223 | --- | 2.5 | C | 0.25 | 0.25 | 0.27 | 253 | --- | 12.5 | C | 0.012 | 0.013 | 0.015 |
| 224 | 316 | 3.0 | A | 0.15 | 0.15 | 0.17 | 254 | 336 | 15.0 | A | 0.009 | 0.009 | 0.010 |
| 225 | 317 | 3.0 | B | 0.15 | 0.15 | 0.17 | 255 | 337 | 15.0 | B | 0.009 | 0.009 | 0.010 |
| 226 | --- | 3.0 | C | 0.15 | 0.15 | 0.17 | 256 | --- | 15.0 | C | 0.009 | 0.009 | 0.010 |
| 227 | 318 | 4.0 | A | 0.10 | 0.10 | 0.12 | 257 | 338 | 20.0 | A | 0.006 | 0.006 | 0.007 |
| 228 | 319 | 4.0 | B | 0.10 | 0.10 | 0.12 | 258 | 339 | 20.0 | B | 0.006 | 0.006 | 0.007 |
| 229 | --- | 4.0 | C | 0.10 | 0.10 | 0.12 | 259 | --- | 20.0 | C | 0.006 | 0.006 | 0.007 |

1/ The corresponding maximum wattage losses, which in no case shall exceed 3.0 watts, may be calculated as $I^2R$ or $I^2Z$.





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TABLE II. Cross reference - superseded part numbers.

| Superseded MIL-C-39019/6B part number M39019/6- | New part number M39019/06- | Superseded MIL-C-39019/6B part number M39019/6- | New part number M39019/06- | Superseded MIL-C-39019/6B part number M39019/6- | New part number M39019/06- |
|---|---|---|---|---|---|
| 1 | 200 | 42 | 231 | 83 | 218 |
| 2 | 201 | 43 | 230 | 84 | 219 |
| 3 | 200 | 44 | 231 | 85 | 218 |
| 4 | 201 | 45 | 230 | 86 | 219 |
| 5 | 200 | 46 | 231 | 87 | 227 |
| 6 | 201 | 47 | 230 | 88 | 228 |
| 7 | 200 | 48 | 231 | 89 | 227 |
| 8 | 201 | 49 | 230 | 90 | 228 |
| 9 | 200 | 50 | 231 | 91 | 227 |
| 10 | 201 | 51 | 248 | 92 | 228 |
| 11 | 209 | 52 | 249 | 93 | 233 |
| 12 | 210 | 53 | 248 | 94 | 234 |
| 13 | 209 | 54 | 249 | 95 | 233 |
| 14 | 210 | 55 | 248 | 96 | 234 |
| 15 | 209 | 56 | 249 | 97 | 233 |
| 16 | 210 | 57 | 248 | 98 | 234 |
| 17 | 209 | 58 | 249 | 99 | 236 |
| 18 | 210 | 59 | 248 | 100 | 237 |
| 19 | 209 | 60 | 249 | 101 | 236 |
| 20 | 210 | 61 | 254 | 102 | 237 |
| 21 | 215 | 62 | 255 | 103 | 236 |
| 22 | 216 | 63 | 254 | 104 | 237 |
| 23 | 215 | 64 | 255 | 105 | 242 |
| 24 | 216 | 65 | 254 | 106 | 243 |
| 25 | 215 | 66 | 255 | 107 | 242 |
| 26 | 216 | 67 | 254 | 108 | 243 |
| 27 | 215 | 68 | 255 | 109 | 242 |
| 28 | 216 | 69 | 254 | 110 | 243 |
| 29 | 215 | 70 | 255 | 111 | 245 |
| 30 | 216 | 71 | 257 | 112 | 246 |
| 31 | 224 | 72 | 258 | 113 | 245 |
| 32 | 225 | 73 | 257 | 114 | 246 |
| 33 | 224 | 74 | 258 | 115 | 245 |
| 34 | 225 | 75 | 257 | 116 | 246 |
| 35 | 224 | 76 | 258 | 117 | 251 |
| 36 | 225 | 77 | 257 | 118 | 252 |
| 37 | 224 | 78 | 258 | 119 | 251 |
| 38 | 225 | 79 | 257 | 120 | 252 |
| 39 | 224 | 80 | 258 | 121 | 251 |
| 40 | 225 | 81 | 218 | 122 | 252 |
| 41 | 230 | 82 | 219 | 123 | 233 |



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Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AV, AT
Navy - AS, MC, SH
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

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